Proof: $(ABC)(C^{-1}B^{-1}A^{-1}) = (AB)(C\cdot(C^{-1}B^{-1}A^{-1}))$ (Theorem 2(a))

$= (AB)(C\cdot(C^{-1}(B^{-1}A^{-1})))$ (Theorem 2(a))

$= (AB)((C\cdot C^{-1})\cdot(B^{-1}A^{-1}))$ (Theorem 2(a))

$= (AB)(I\cdot(B^{-1}A^{-1}))$ (By Defn.)

$= (AB)(B^{-1}A^{-1})$ (Theorem 2(e))

$= A(B\cdot(B^{-1}A^{-1}))$ (Theorem 2(a))

$= A((B\cdot B^{-1})\cdot A^{-1})$ (Theorem 2(a))

$= A(I\cdot A^{-1})$ (By Defn.) $= AA^{-1}$ (Theorem 2(e)) $= I$ (By Defn.)

$(C^{-1}B^{-1}A^{-1})(ABC) = (C^{-1}B^{-1})(A^{-1}(ABC))$ (Theorem 2(a))

$= (C^{-1}B^{-1})(A^{-1}(A(BC)))$ (Theorem 2(a))

$= (C^{-1}B^{-1})((A^{-1}A)(BC))$ (Theorem 2(a))

$= (C^{-1}B^{-1})(I(BC))$ (By defn.)

$= (C^{-1}B^{-1})(BC)$ (Theorem 2(e))

$= C^{-1}(B^{-1}(BC))$ (Theorem 2(a))

$= C^{-1}((B^{-1}B)C)$ (Theorem 2(a))

$= C^{-1}(IC)$ (By defn.) $= C^{-1}C$ (Theorem 2(e)) $= I$ (By defn. of inverse) $\square$

- Now we shall prove that the matrix D for both the equations is unique.

$(ABC)D = I \Rightarrow A^{-1}((ABC)D) = A^{-1}I \Rightarrow (A^{-1}(ABC))D = A^{-1}I$ (Theorem 2(a))

$\Rightarrow (A^{-1}(A(BC)))D = A^{-1}I$ (Theorem 2(a))

$\Rightarrow ((A^{-1}A)(BC))D = A^{-1}I$ (Theorem 2(a)) $\Rightarrow (I(BC))D = A^{-1}I$ (By defn. of inverse)

$\Rightarrow (BC)D = A^{-1}$ (Theorem 2(e)) $\Rightarrow B^{-1}(BCD) = B^{-1}A^{-1}$

$\Rightarrow B^{-1}(B(CD)) = B^{-1}A^{-1}$ (Theorem 2(a)) $\Rightarrow (B^{-1}B)(CD) = B^{-1}A^{-1}$ (Theorem 2(a))

$\Rightarrow I(CD) = B^{-1}A^{-1}$ (By defn. of inverse) $\Rightarrow CD = B^{-1}A^{-1}$ (Theorem 2(e))

$\Rightarrow C^{-1}(CD) = C^{-1}(B^{-1}A^{-1}) \Rightarrow (C^{-1}C)D = C^{-1}B^{-1}A^{-1}$ (Theorem 2(a))

$\Rightarrow ID = C^{-1}B^{-1}A^{-1}$ (By defn. of inverse) $\Rightarrow \boxed{D = C^{-1}B^{-1}A^{-1}}$ (Theorem 2(e))

$D(ABC) = I \Rightarrow (D(ABC))C^{-1} = I\cdot C^{-1} \Rightarrow ((D(AB))C)C^{-1} = IC^{-1}$ (Theorem 2(a))

$\Rightarrow (D(AB))(CC^{-1}) = IC^{-1}$ (Theorem 2(a))

$\Rightarrow (D(AB))I = IC^{-1}$ (By Defn. of inverse)